Let us take a look at some of the biggest 'software development horrors' that have brought researchers, corporations and sometimes even countries to their knees.

Code can crash and burn

Mars orbiter crashed by metrics (1999):

In September of 1999 the mars climate orbiter crashed due to a technical snag on its ground-based navigation software. The Mars Climate Orbiter (MCO) was to observe Martian weather systems and relay communications. The spacecraft was to fire its engines to enter orbit at an altitude of 145 km. Instead, the spacecraft entered the atmosphere at 57 km, and the friction caused it to burn up. This was no computer failure, but a human using the wrong units. NASA conventionally uses metric units, but USA-based engineers often use imperial. To put the orbiter in the proper orbit, a calculation for its momentum (mass x velocity) was needed.

Instead of the expected metric Newton-seconds, an engineer in the organization used imperial pound-seconds. The resultant calculation was 4.45 times bigger than it should have been. This caused the orbiter to enter at a much lower orbit than expected and the additional friction at the lower orbit was enough to compromise the structure of the craft and fry its systems. This project had cost NASA a cumulative of over $100 million.

AT&T communication down across USA (1990):

At 2:25pm on Monday, 15th January 1990, network managers at AT&T's Network Operations Center in Bedminster, N.J. began noticing an alarming number of red warning signals from various parts of their world-wide network.

Working backwards through the data, technicians identified the problem. The New York switch had performed a routine self-test that indicated it was nearing its load limits. As standard procedure, the switch performed a four-second maintenance reset and sent a message over the signalling network that it would take no more calls until further notice. After reset, the New York switch began to distribute the signals that had backed up during the time it was off-line.